

Project Title
**RouteOpt: A Data-Cleaning and Approximation Engine for
Capacitated Multi-Vehicle Route Optimization with Time
Windows**

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Logistics operators must plan daily routes visiting many stops at minimum total distance and time while respecting vehicle capacity and delivery time windows. This problem, a variant of the Capacitated Vehicle Routing Problem with Time Windows (CVRPTW), is NP-hard and exactly solvable only for small instances, so real-world fleets require fast approximation methods. This project presents RouteOpt, an engine that ingests raw stop and address data, cleans and geocodes it, and produces near-optimal per-vehicle routes. The pipeline standardizes messy addresses, deduplicates stops, clusters and assigns them to vehicles under capacity constraints, and applies construction heuristics (nearest-neighbor, savings algorithm) followed by local-search improvement (2-opt, Or-opt) to minimize travel cost while honoring time windows. The system reports total distance, estimated travel time, vehicle utilization, and time-window violations for each route, giving operators a measurable trade-off between solution quality and computation time compared to exact solvers.

Introduction

The Vehicle Routing Problem (VRP), first introduced by Dantzig and Ramser, generalizes the Traveling Salesman Problem to multiple vehicles and is a foundational problem in combinatorial optimization and logistics. Practical variants add vehicle capacity limits (CVRP) and customer-specific delivery windows (CVRPTW), both highly relevant to same-day delivery, courier, and field-service operations. Exact methods such as branch-and-bound, branch-and-cut, and integer linear programming guarantee optimal solutions but scale poorly, becoming computationally intractable beyond a few dozen stops. Consequently, industry systems rely on constructive heuristics (nearest-neighbor, Clarke-Wright savings) and metaheuristic local search (2-opt, Or-opt, simulated annealing, tabu search) to obtain near-optimal routes in seconds rather than hours. A separate but equally critical challenge is data quality: real-world address and stop data is frequently inconsistent, duplicated, incomplete, or inaccurately geocoded, and existing routing tools often assume clean coordinates as input, leaving data preparation as a manual, error-prone step. This project addresses both gaps by building an end-to-end engine that first cleans and standardizes raw stop data, then assigns stops to vehicles and computes near-optimal routes, and finally reports cost and quality metrics so operators can evaluate the trade-off between speed and optimality.

Software and hardware requirements

Software: Python 3.10+; libraries — pandas and NumPy (data cleaning), scikit-learn or OR-Tools (clustering and optimization routines), Geopy/Nominatim or Google Maps Geocoding API (address geocoding), NetworkX (graph and distance-matrix handling), Folium or Plotly (route visualization), Flask/FastAPI (optional web interface); VS Code or PyCharm as IDE; Git for version control. Hardware: standard development machine with a multi-core CPU (Intel i5/Ryzen 5 or higher), minimum 8 GB RAM (16 GB recommended for large instances), 10 GB free disk space, and an internet connection for geocoding API calls.

Signature of the Guide

Course Instructor